ECON100B - SECTION 5

Cobb Douglas Pt. 2 and PSET Practice

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 1 due Friday, 2/13

Agenda

- Cobb Douglas PSET problems
- Solow Model (time permitting)

Question of the Day

1. Consider the classic Cobb Douglas function setup: $F(K, L) = K^{1/3}L^{2/3}$. What happens to output when we capital double and labor doubles? What happens to output when capital doubles, but labor stays constant? What happens to output when labor doubles, but capital stays constant?

2. What is an album/song/artist that you’ve been listening to recently?

The Cobb Douglas Model Pt. 2 & Intro to Solow Model

Problem Set Questions

(e) [2 points] In Chapter 4, we will see that GDP is modeled well by this Cobb-Douglas production function:

$$Y = A K^{\alpha} L^{1-\alpha} \qquad (1.3)$$

where α is a constant number between 0 and 1 that is also capital's share of income. In the textbook, $\alpha = 1/3$, which is an appropriate level of the parameter for U.S. data.

Take the growth rate of both sides of equation (1.3), treating A, K, and L as variables that potentially have growth rates. Simplify and briefly describe your answer:

1. $z = x / y$, then $g_z = g_x - g_y$.
2. If $z = x \times y$, then $g_z = g_x + g_y$.
3. If $z = x^a$, then $g_z = a \times g_x$.

Problem Set Questions

(f) [2 points] Using your answer above, describe what happens to Y if A is constant while K and L both increase by 2%. Be specific and briefly explain.

(g) [2 points] Using your answer above, describe what happens to Y if A grows by 3% while K and L both increase by 2%. Be specific and briefly explain.

Problem Set Questions

2. [12 points total] Cobb-Douglas Come Out to Play?

Although there are some phenomena in economics that the Cobb-Douglas function cannot capture realistically, it is a remarkably versatile tool that you will likely see again and again in your undergraduate studies. Recall that in Chapter 4, we see that the U.S. macroeconomy can be modeled well using Cobb-Douglas production:

$$Y = A\,K^{\alpha}\,L^{1-\alpha} \qquad (2.1)$$

where $\alpha = 1/3$ is capital's share of income. When we take first partial derivatives of equation (2.1) we reveal the useful functional forms of the marginal products of capital (MPK) and of labor (MPL), which in competitive markets equal their factor prices: the interest rate r and the wage w:

$$\frac{\partial Y}{\partial K} = MPK = r \qquad r = \frac{1}{3}A\left(\frac{L}{K}\right)^{2/3} \qquad (2.2)$$

$$\frac{\partial Y}{\partial L} = MPL = w \qquad w = \frac{2}{3}A\left(\frac{K}{L}\right)^{1/3} \qquad (2.3)$$

(a) [2 points] Take the growth rate of both sides of equation (2.2), which shows r as a function of A, K, and L, and simplify. If A is constant, then what must be true about the growth rate of r when labor and capital are growing at the same rate and thus the capital-labor ratio is constant? Show your work.

Problem Set Questions

(b) [2 points] Using your answer above, what happens to r if there is no growth in A but K rises above L by 1%?

Problem Set Questions

Consider what happens when we take the natural log of both sides of equation (2.3), which is similar to taking the growth rate:

$$\log w = \log\frac{2}{3} + \log A + \frac{1}{3}\log K - \frac{1}{3}\log L \qquad (2.4)$$

(c) [3 points] In the **labor market**, assume technology A and capital K are both fixed. On the axes below, graph the simple relationship that you see in equation (2.4) between the log wage on the vertical axis and $\log L$ on the horizontal axis. Explain what you drew. Report the numerical values of the vertical intercept and the slope.

(d) [2 points] In the picture you just drew, add a vertical **labor supply curve** and show the equilibrium wage and employment in the labor market. Briefly explain below.

BIG PICTURE QUESTIONS

- **What sustains growth in the long run?**
  - Will it continue in the future?
- **Why are some countries so far behind the frontier?**
  - What might help them close the gap?

FOOD FOR THOUGHT

If I put you in charge of every aspect of a country right now and told you your only goal was to grow the economy's per-capita income, what would you focus on?

IS CAPITAL ACCUMULATION KEY TO GROWTH?

- Seems plausible - just keep investing in more productive capacity
- Conventional wisdom in the 1950s - Yes!
- **Solow (1956) - No, you can’t grow indefinitely with capital accumulation alone**

Solow Model Setup

- Consider a large farm that grows a single crop, corn
- ¾ of the corn is reserved for eating, while ¼ of the corn is saved for planting for next year
- Each saved corn kernel produces ten ears of corn, so the size of harvest can grow larger in every year
- The Solow model helps us describe and understand capital accumulation ("corn") over time

The Production Function, Revisited

$$Y_t = F\left(K_t, L_t\right) = \bar{A}K_t^{1/3}L_t^{2/3}$$

- We add a new time subscript to the production function
- In the model economy, output can be used for one of two purposes, consumption or investment:

$$C_t + I_t = Y_t,$$

- This is a resource constraint, as the economy is constrained to spending on Consumption or Investment, equal to output

Accumulation of Capital

- We saw previously, that we saved a certain amount in investment

$$C_t + I_t = Y_t,$$

- Our future capital is determined by the following equation:

$$K_{t+1} = K_t + I_t - \bar{d}K_t.$$

- What this equation says: capital next year is equal to capital this year added to investment this year, minus the depreciation of capital
- We use an exogenous depreciation rate, commonly 7% or 10%

Accumulation of Capital

Time, t	Capital, K_t	Investment, I_t	Depreciation, $\bar{d}K_t$	Change in capital, ΔK_{t+1}
0	1,000	200	100	100
1	1,100	200	110	90
2	1,190	200	119	81
3	1,271	200	127	73
4	1,344	200	134	66
5	1,410	200	141	59

How do we determine investment?

- **Investment:** $S = I = sY$
- **Consumption:** $C = (1 - s)Y$

- We set "s" as the fraction of our output that is set aside for investment
- "1-s" represents the fraction of our output that is set aside for consumption
- By logic, s + (1 – s) has to be equal to 1, where s can be any number between 0 and 1

The real interest rate

- The real interest rate is the amount that a person can earn by saving one unit of output for a year, or equivalently the amount a person must pay to borrow one unit of output for a year
- The real interest rate equals the rental price of capital that clears the capital market, which in turn is equal to the marginal product of capital
- Why? Savings (“S”) = Investment (“I”) which becomes the capital supplied in the economy

Output per person

Recall from the Cobb Douglas model:

Output per person as:

$$y_t = Y_t/L_t$$

Capital per person:

$$k_t = K_t/L_t$$

$$\begin{aligned} y_t \equiv \frac{Y_t}{L_t} &= \frac{\bar{A}K_t^{1/3}L_t^{2/3}}{L_t} \\ &= \frac{\bar{A}K_t^{1/3}}{L_t^{1/3}} \\ &= \bar{A}k_t^{1/3}. \end{aligned}$$

Output per person

- Solving for output per person tells us that output per person is dependent on capital
- To understand how output per person (y_t) changes over time, we need to understand how capital per person (k_t) changes over time

$$\underbrace{\Delta K_{t+1}}_{\text{change in capital}} = \underbrace{\bar{s}Y_t - \bar{d}K_t}_{\text{net investment}}.$$

- The change in capital stock is equal to investment minus depreciation, a.k.a net investment

$$\underbrace{\Delta k_{t+1}}_{\text{change in capital per person}} = \underbrace{\bar{s}y_t - \bar{d}k_t}_{\text{net investment per person}}.$$

The Solow Model

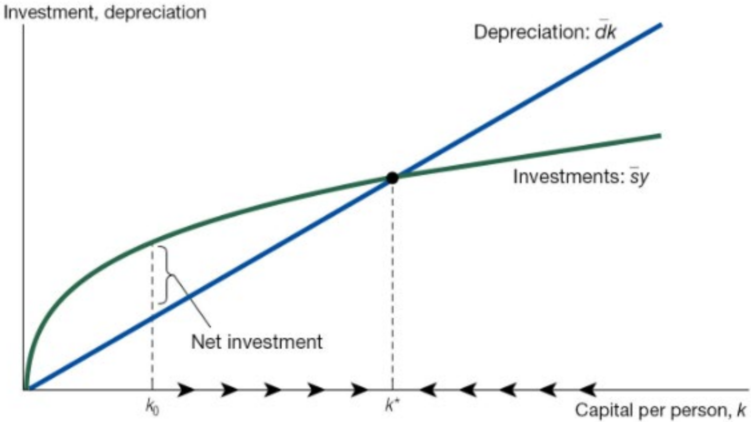
- The essence of the Solow model is understanding of output per person changes over time (dependent on capital) and how capital changes over time:

$$y_t = \bar{A}k_t^{1/3}, \tag{5.6}$$

$$\Delta k_{t+1} = \bar{s}y_t - \bar{d}k_t. \tag{5.7}$$

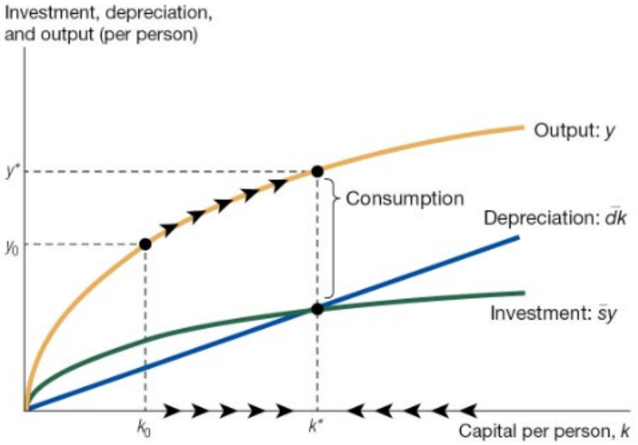
- We cannot solve these equations algebraically (too many endogenous variables) but can analyze it in a graph

The Solow Diagram



- Where our investment curve can be written as $\bar{s}y = \bar{s}\bar{A}k^{1/3}$.

The Solow diagram with output



The Steady state

- While we can't solve for capital at any given point, we are able to solve for the steady state level of capital:

$$\bar{s}y^* = \bar{d}k^*.$$

$$\bar{s}\bar{A}k^{*1/3} = \bar{d}k^*.$$

$$k^* = \left(\frac{\bar{s}\bar{A}}{\bar{d}}\right)^{3/2}.$$

Steady State Output

- Associated with a steady-state level of capital is a steady state level of production

$$y^* = \bar{A}k^{*1/3}.$$

- Thus, we achieve the following expression for steady-state production if we plug in k* into the equation:

$$y^* = \left(\frac{\bar{s}}{\bar{d}}\right)^{1/2} \bar{A}^{3/2}.$$